IS_NOT_FUTURE}. Following this framework, we estimate vector representations for each feature value using the variant of LDA proposed by Park et al. (2024a). The objective is to find a vector that reduces within-class variation and highlights differences to all other classes. Unlike standard LDA, which relies on both within-class and between-class covariance, this variant omits the latter, enabling the computation of each vector independently of the others.

Formally, for each binary feature w, we compute a **normalized class direction** $\tilde{h}_w$ from the empirical mean of the class-specific activations $\mathbb{E}(h_w)$, adjusted by the pseudo-inverse of the class covariance $\mathrm{Cov}(h_w)^\dagger$:

$$\tilde{h}_w = \frac{\mathrm{Cov}(h_w)^\dagger \mathbb{E}(h_w)}{\|\mathrm{Cov}(h_w)^\dagger \mathbb{E}(h_w)\|_2}. \quad (2)$$

This unit vector captures the direction of the class in residual space. To incorporate the strength of the signal, we scale the direction by the projection of the class mean onto it:

$$\bar{\ell}_w = (\tilde{h}_w^\top \mathbb{E}(h_w))\tilde{h}_w. \quad (3)$$

The resulting vector $\bar{\ell}_w$ encodes the orientation and intensity of the concept in activation space.

Importantly, because each vector is computed independently, the method avoids enforcing any pre-defined class structure of tense and aspect. Instead, the representational geometry that emerges reflects the structure learned by the model itself.

Another key concept introduced by Park et al. (2024a) is **binary contrast**, which captures the distinction between two categorical values within the same parent category and is computed as the vector difference between their feature vectors. In our analysis, we use binary contrasts to model categories in a lower-dimensional space. This allows us to (i) compare subordinate features within a category where their number exceeds the available representational dimensions (i.e., for aspectual values), and (ii) approximate latent dimensions for cross-category comparisons between tense and aspect.

3.2 Results

Representations of tense and aspect emerge early in the model. Our probing classifiers predict tense and aspect from the embedding layer with f1-scores above 90% and improve further with model depth, especially for the fine-grained tense-aspect combinations, see Table 1.

	Tense	Aspect	Tense-Aspect
Llama-8B	1.0	0.98	0.93
Qwen-7B	1.0	0.98	0.92

Table 1: Target-wise F1-scores from the best-performing probe across layers.

This demonstrates that contextualization is beneficial, as many tenses and aspects are expressed across multiple tokens. For detailed results across all layers and more aggregation strategies, see App. E. The findings are consistent across targets and models and similar to earlier work that suggests syntactic processing happens before more complex semantic processing (He et al., 2024).

Grammatical properties form subspaces in representation space. To analyze the LDA results, we use the 2D and 3D visualizations of Park et al. (2024a). To assess whether hidden representations encode a separation between categories of a single grammatical feature, we project test set embeddings from Qwen-7B onto selected directions.

For tense, we use $\bar{\ell}_{\text{PRESENT}}$, $\bar{\ell}_{\text{PAST}}$ and $\bar{\ell}_{\text{FUTURE}}$ as projection axes. For aspect, we use binary contrasts retrieved from vector differences to represent the four categories: $\bar{\ell}_{\text{PROGRESSIVE}} - \bar{\ell}_{\text{SIMPLE}}$, $\bar{\ell}_{\text{PERFECT}} - \bar{\ell}_{\text{SIMPLE}}$ and $\bar{\ell}_{\text{PERFECT PROGRESSIVE}} - \bar{\ell}_{\text{SIMPLE}}$.

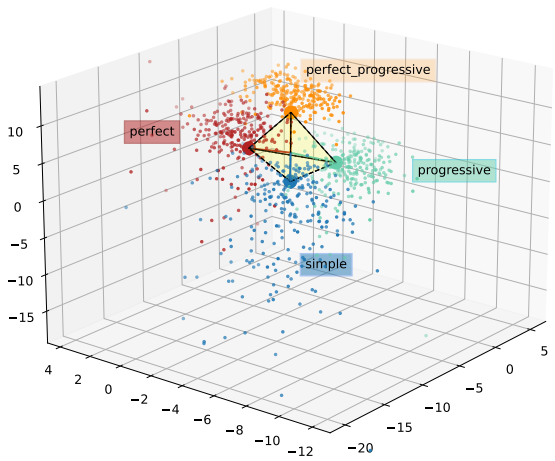

Figure 2: Projection of Qwen-7B hidden states (L0) along LDA-based **aspect** directions.

In both cases, 3D projections reveal distinct clusters corresponding to the underlying grammatical categories: three well-separated clusters for tense

(Figure 1, explained variance: 0.72) and four for aspect (Figure 2, explained variance: 0.70). They span a convex region in both cases, suggesting that the feature vectors define structured subspaces. See App. F for additional cluster quality metrics of both models.

Tense and aspect exhibit representational independence. Next, we examine the relationship between the two grammatical categories by projecting representations onto their respective latent dimensions. Specifically, we use the binary contrasts $\bar{\ell}_{\text{TENSE}} = \bar{\ell}_{\text{FUTURE}} - \bar{\ell}_{\text{PAST}}$ and $\bar{\ell}_{\text{ASPECT}} = \bar{\ell}_{\text{PROGRESSIVE}} - \bar{\ell}_{\text{PERFECT}}$ as proxies for the tense and aspect dimensions[2].

Figure 3 shows the projection of data points onto the inferred tense (x-axis) and aspect (y-axis) dimensions. Similar to earlier feature-wise visualizations, the points cluster according to their grammatical categories. The groups are organized near-orthogonally in latent space, reflecting a clear separation between tense and aspect. This is further supported by the near-zero cosine similarity of 0.02 between the vectors $\bar{\ell}_{\text{TENSE}}$ and $\bar{\ell}_{\text{ASPECT}}$ (see App. F).

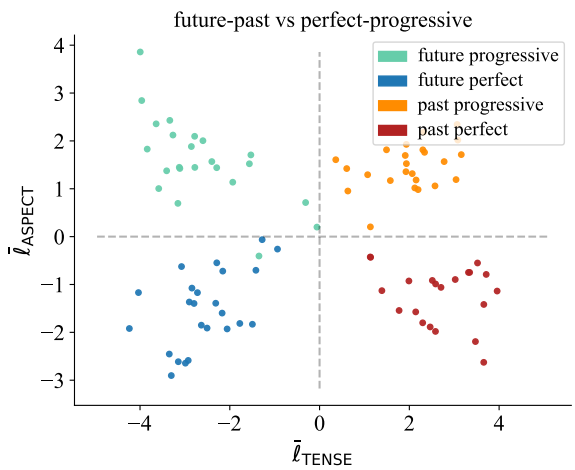

Figure 3: Projection of Llama-8B hidden states (L7) along $\bar{\ell}_{\text{TENSE}}$ and $\bar{\ell}_{\text{ASPECT}}$ clearly separate the data points according to their grammatical property.

4 Multi-Token Steering

After identifying correlational evidence for tense and aspect directions in model representations, we test their causal impact on model behavior via targeted interventions during text generation. We evaluate functional selectivity by checking whether the manipulated outputs express the steered verb property while preserving other verb features and the original meaning. This quantitative analysis is followed by a qualitative study in which we manually examine model outputs and investigate the impact of steering location, strength and duration.

4.1 Experimental Setup

Tasks. We consider three complementary tasks (prompt details in App. G):

1. **Random Sentence Task:** The model is prompted to generate an open-ended sentence, testing whether grammatical concepts can be induced in semantically unconstrained settings.

2. **Repetition Task:** In a few-shot setup, the model must repeat a sentence after two example repetitions. This copying task requires interventions to override contextual information. It evaluates the ability to steer generation when the model's default behavior follows pattern induction.

3. **Temporal Translation Task:** We use a similar few-shot setup where the model must "translate" a sentence into a different tense or aspect. Unlike repetition, this requires internal transformation, allowing us to test whether interventions can influence more complex linguistic transformations.

Steering Methods. We perform steering at a single transformer layer l on the final position $i = -1$ of the input sequence at every generation step. Concretely, we update the residual stream activation vector $h_i^l \in \mathbb{R}^d$ by adding and/or subtracting the normalized LDA-derived concept directions from earlier. These unit vectors correspond to specific tense and aspect values. A scalar steering factor $\alpha \in \mathbb{R}$ scales the strength of the modification. We evaluate three distinct steering strategies across all layers and different α values:

1. **Target-Addition Only (TA):** The standard steering approach which is commonly used in related work such as Rimsky et al. (2024), directly adds the normalized target direction $\bar{\ell}_T$ to the current activation:

$$h^{\text{steered}} = h_i^l + \alpha \bar{\ell}_T. \tag{4}$$

2. **Target-Addition with Source-Subtraction (TA+SS):** To simultaneously steer the target

[2]Any binary contrast among subcategories (e.g., FUTURE-PAST, PRESENT-PAST, FUTURE-PRESENT) lies in the same one-dimensional parent-contrast subspace. The vector differences in Fig. 3 were chosen as examples.

concept and suppress a known source concept $\bar{\ell}_S$, we introduce source-subtraction, which subtracts the source direction with equal weight. This is particularly useful for non-binary features, where source and target are not simply inverses:

$$h^{\text{steered}} = h_i^l + \alpha\bar{\ell}_T - \alpha\bar{\ell}_S. \quad (5)$$

3. **Target-Addition with Projection Subtraction (TA+Proj-SS):** Instead of subtracting the full source vector, this method removes only the component of the activation that lies along $\bar{\ell}_S$. This is achieved by computing and subtracting the projection of h_i^l onto $\bar{\ell}_S$:

$$h^{\text{steered}} = h_i^l + \alpha\bar{\ell}_T - (h_i^l \cdot \bar{\ell}_S)\bar{\ell}_S. \quad (6)$$

To ensure comparability between steered and original generations, we use greedy decoding, where the most likely token is selected at each step.

Evaluation Metrics. We evaluate steering success by giving the generated outputs to the model in a new forward pass without any interventions, extracting their representations and applying the trained probing classifiers, following Brinkmann et al. (2025). We probe not only for the steering target but also for the respective other property. To quantify the effect of our interventions, we define the following four performance metrics:

$$\text{Steering Success} = \frac{|S|}{N},$$
$$\text{Degenerate Rate} = \frac{|D|}{N},$$
$$\text{Efficacy} = \frac{|S \setminus D|}{N},$$
$$\text{Selectivity} = \frac{|S_F \setminus D|}{N}.$$

Here, N is the number of test samples[3], $S \subseteq \{1, \ldots, N\}$ is the set of successfully steered samples and $D \subseteq \{1, \ldots, N\}$ the set of degenerate outputs. An output is considered degenerate, if it either (i) forms an incomplete sentence by missing a verb, as detected by a part-of-speech (POS) tagger, or (ii) exhibits excessive n-gram repetition or low n-gram diversity (see App. I for thresholds). S_F is the subset of S for which the probe's label for the not steered property stays constant. Finally, we report the relative change in perplexity to measure the impact of steering on fluency and coherence.

[3] N is task dependent, see App. G.

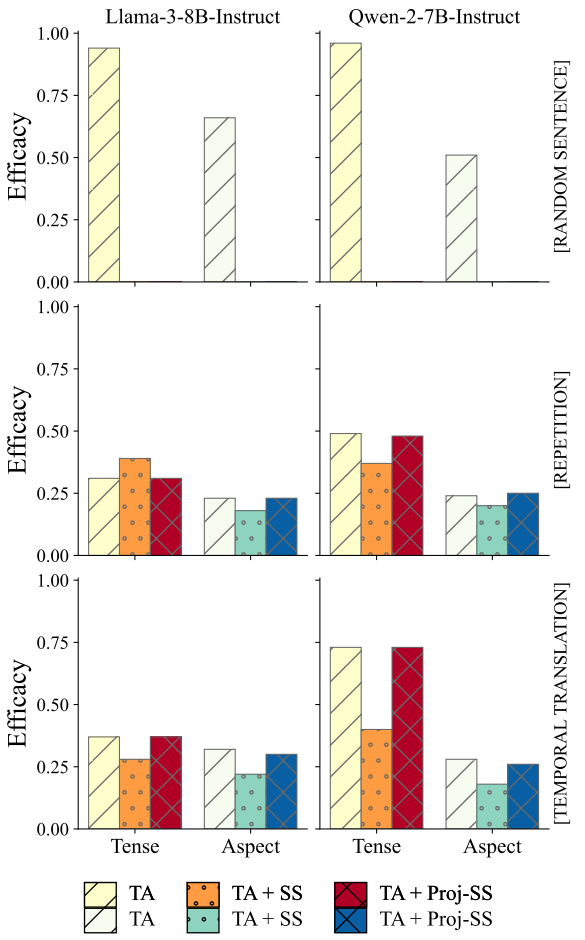

Figure 4: Efficacy of different steering methods. TA+SS and TA+Proj-SS are not applied for the Random Sentence Task because there is no source feature direction to be subtracted. Both models show similar trends with tense being easier to steer than aspect, and random sentences easier than few-shot tasks.

4.2 Quantitative Results

For each task and steering method, we perform a grid search over all layers and selected α values (see App. H) and report the configuration that yields the highest efficacy (Figure 4).

Steering success varies widely by task and target. Overall, steering tense achieves substantially higher success than aspect. For example, on the random sentence task, efficacy is near-perfect for tense (94% for Llama-8B, 96% for Qwen-7B) but noticeably lower for aspect (66% for Llama-8B, 51% for Qwen-7B). Steering in the few-shot settings reduces performance for both targets, with the best scenario reaching 73% (Qwen-7B, tense) and the worst just 18% (aspect for both models). Steering tense or aspect is significantly more difficult when the task requires conflicting verb properties to